

Portfolio 1

```
library(ggplot2)
library(readr)
library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

`filter`, `lag`

The following objects are masked from 'package:base':

`intersect`, `setdiff`, `setequal`, `union`

```
library(tidyr)
```

The dataset I am investigating is focused on student Performance in Exams, found on kaggle: <https://www.kaggle.com/datasets/spscientist/students-performance-in-exams>. The data directly states in its acknowledgements “http://roycekimmons.com/tools/generated_data/exams” which includes “generated data.” To understand what this looks like in other dataset I am going to look for the exact signals and tells that can further prove that the data is fabricated. The dataset claims to be looking into “marks secured by the students high school students from the United States.” One interesting thing to note is that it says how the marks were “found” while being a generated dataset.

Some questions I am interested in to understand what parts of the data seems generated versus which parts might have passed as a real dataset are as follows:

1. Do the exams scores have distributions that would be found in real life? Any unusual clusters or groups?

2. Did the students who studied more score better on the tests?
3. Are there students that have the same records?
4. Do the writing and reading exam scores correlate?

```
exams <- read_csv("Exam Data /StudentsPerformance.csv")
```

```
Rows: 1000 Columns: 8
```

```
-- Column specification -----
```

```
Delimiter: ","
```

```
chr (5): gender, race/ethnicity, parental level of education, lunch, test pr...
```

```
dbl (3): math score, reading score, writing score
```

i Use ``spec()`` to retrieve the full column specification for this data.

i Specify the column types or set ``show_col_types = FALSE`` to quiet this message.

```
scores <- exams |>
```

```
  pivot_longer(
```

```
    cols = c(`math score`, `reading score`, `writing score`),
```

```
    names_to = "subject",
```

```
    values_to = "score"
```

```
  )
```

```
ggplot(scores, aes(x = score)) +
```

```
  geom_histogram(binwidth = 5) +
```

```
  facet_wrap(~subject) +
```

```
  labs( y = "Number of Students", x = "Exam Score", title = "Diagram One: Exam Scores Distribution")
```

Diagram One: Exam Scores Distribution by Subject

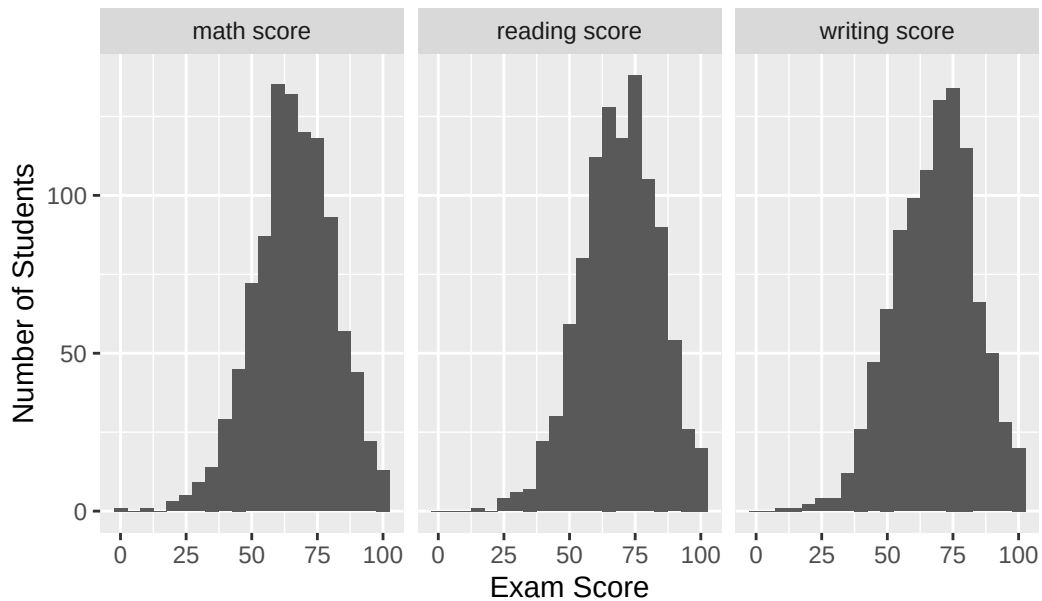


Diagram one displays the exam score distributions for math, reading, and writing. There are a few things that may not reflect a dataset that was collected from real students. One, all classes have the same amount of students exactly, whereas in reality you would expect some fluctuation. Two, there was a student who received a score of 0 in math which seems very unlikely occur. Overall, to answer question one, the distributions seem fairly normal at first glance but after some consideration there are some things that seem unrealistic. Ultimately, this supports the idea that this data was generated not genuinely collected.

Looking at question two, it may be interesting to compare math scores with those who took the preparation course with those students who did not.

```
ggplot(exams, aes(x = `test preparation course`, y = `math score`)) +  
  geom_boxplot() +  
  theme_minimal() +  
  labs(  
    x = "Test Preparation Course",  
    y = "Math Exam Score",  
    title = "Diagram Two: Math Exam Score and Test Preparation Course")
```

Diagram Two: Math Exam Score and Test Preparation Course

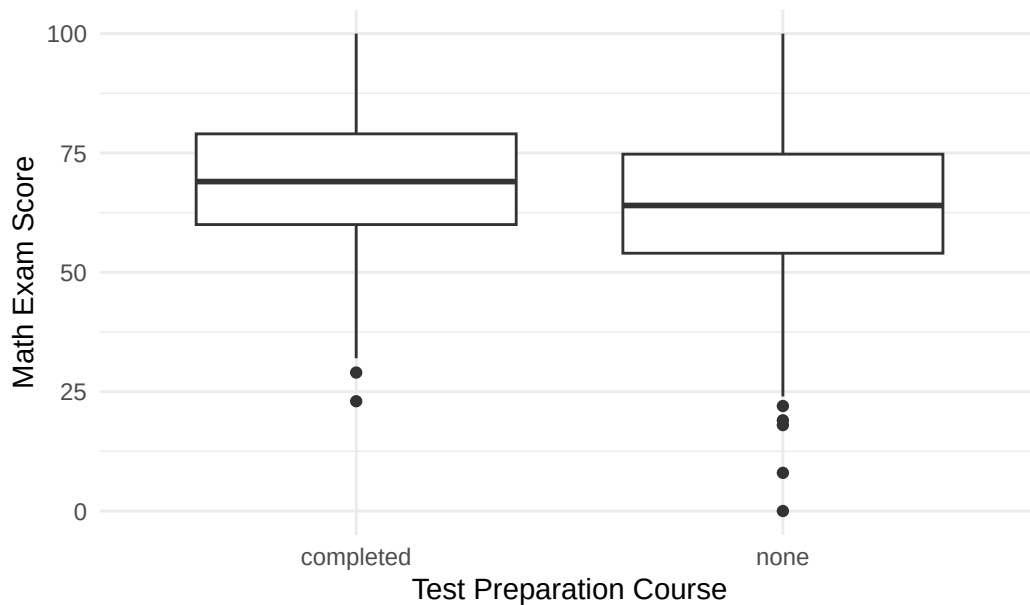


Diagram two clearly illustrates that the students who took the test preparation course did not do much better than those who did take the course. This is not what would be expected in reality, it would make much more sense if the students who took the preparation course achieved significantly higher scores. However, it could be the case that the course was not very well designed and thus was less impactful on exam scores. One thing that would suggest the data is real is the outliers in the “none” category (the students who did not take the preparation course) have lower scores, but it is more important to look at the data as a whole.

```
exams %>%
  count(`math score`, `reading score`, `writing score`) %>%
  filter(n > 1)
```

```
# A tibble: 20 x 4
  `math score` `reading score` `writing score`     n
    <dbl>         <dbl>         <dbl> <int>
1         49           58           60         2
2         53           54           48         2
3         55           65           62         2
4         61           72           70         2
5         62           66           68         2
6         65           77           74         3
7         65           81           81         2
8         66           74           81         2
```

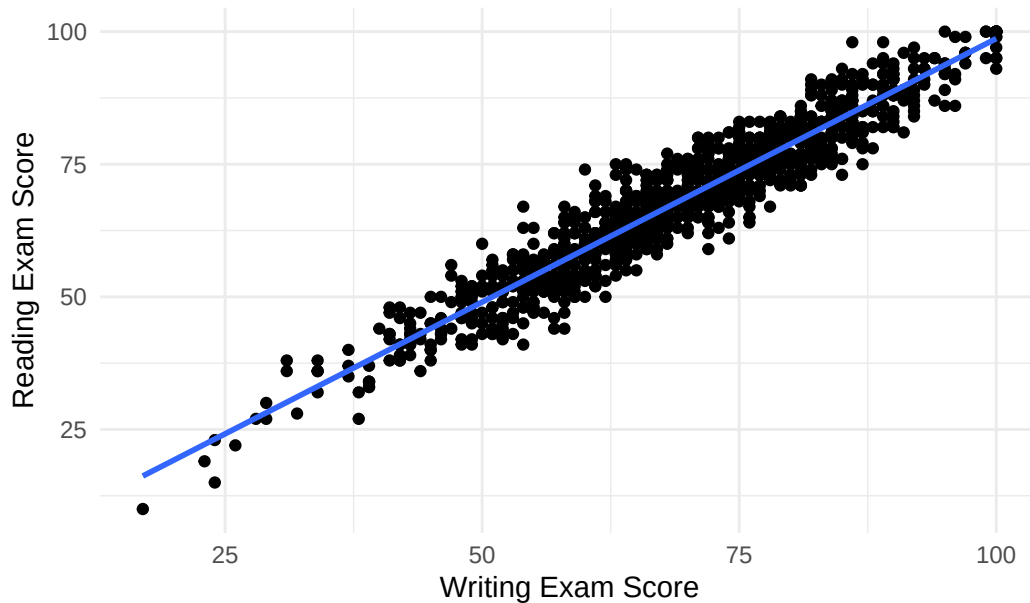
9	67	86	83	2
10	68	70	66	2
11	69	77	77	2
12	71	74	68	2
13	73	68	66	2
14	73	76	78	2
15	74	74	72	2
16	75	88	85	2
17	82	82	80	2
18	82	93	93	2
19	92	100	100	2
20	100	100	100	3

The table above shows the duplicates of exam scores accross the three subjects. While it is not surprising that there are duplicates (because there are 1000 students being studied) there are 20 combinations of scores that were repeated by two or three students. Another potentially unexpected statistic is there are three students that got 100% on all three subjects, while this does not inherently make the data synthetic, it is worth noting. Overall, this suggests but does not directly prove or confirm the data is fabricated.

```
ggplot(exams, aes(x = `reading score`, y = `writing score`)) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE) +
  labs(
    title = "Diagram Three: Relationship Between Reading and Writing Scores",
    x = "Writing Exam Score",
    y = "Reading Exam Score"
  ) +
  theme_minimal()
```

`geom\_smooth()` using formula = 'y ~ x'

Diagram Three: Relationship Between Reading and Writing Sc



```
cor(exams$`reading score`, exams$`writing score`)
```

```
[1] 0.9545981
```

Diagram three displays a remarkably strong positive correlation between reading and writing exam scores. The exact correlation is 0.955, which is very strong. Overall, this is very strong but could be argued to be a reasonable finding since reading and writing are similar subjects that rely on a similar skill set.

Overall, after exploring questions 1-4 it can be determined that the data is not guaranteed to be fabricated but it does have several suspicious and unrealistic set ups that the answers to these questions displayed, so I think it is likely synthetic. To be prove the data is real, it would be interesting to know which school(s) this data was collected from. Additionally, if it was stated how the data was collected. For example, exam scores make sense to just have been collected by the professors but it could have been a survey that students filled out.